Ordering PixelPine rev A at JLCPCB

Generated 2026-10-02 from `hardware/pixelpine.kicad_pcb` **as saved**, hand edits included (GND pour on both layers, extra vias, rotated test pads). Two orders: the tree with assembly, and the stand fin as a bare PCB.

File	Use
`pixelpine_tree_gerbers.zip`	**Order 1**: the tree, PCB + assembly (Gerbers + Excellon drill)
`jlc_bom.csv`, `jlc_cpl.csv`	**Order 1**, assembly step: bill of materials and placement file
`pixelpine_stand_gerbers.zip`	**Order 2**: the stand fin, PCB only (no parts, no holes)
`preview_tree_top/bottom.png`, `preview_stand_top/bottom.png`	3D renders of exactly what went into the Gerbers
`gerbers_tree/`, `gerbers_stand/`	the same files unzipped, for a Gerber viewer

Regenerate after editing the board in KiCad (it doesn't change the board):

```
"/c/Program Files/KiCad/10.0/bin/python.exe" hardware/scripts/export_gerbers.py
```

```
python hardware/scripts/export_fab.py
```

Why two orders: the fin is a separate piece in the KiCad file, and JLC counts every piece that can be separated (milled, V-cut or tabbed) as an extra design, which costs extra (JLC: what counts as different designs). If you'd rather have one order, upload the tree zip with *Different design = 1* and order the fin separately later. Both pieces must be **1.6 mm** thick: the cross-lap slots are 1.7 mm wide.

Order 1: the tree (PCB + assembly)

Upload `pixelpine_tree_gerbers.zip` on jlcpcb.com → *Order now*. Settings (everything not listed: default):

Setting	Value	Why
Base material / Layers	FR-4 / **2**	
Dimensions	50.53 × 70.00 mm	read from the Gerber
Different design	1	
Delivery format	Single PCB	
PCB thickness	**1.6 mm**	the stand fin's slot is cut for 1.6 mm
PCB color / Silkscreen	Green / White	
Surface finish	**LeadFree HASL**	RoHS (compliance plan)
Outer copper weight	1 oz	
Via covering	Tented	
Min via hole size / diameter	0.3 mm / (0.4/0.45 mm)	the board uses 0.3 mm drills, 0.4 mm pads: no surcharge
Board outline tolerance	±0.2 mm (regular)	
Mark on PCB	**Remove Mark**	otherwise JLC prints its order number wherever it likes; the board has no marker for "specify position" (JLC ordering notes)
Electrical test	Flying probe	

Then switch on **PCB Assembly**:

Setting	Value
PCBA type	Economic
Assembly side	Top side

Setting	Value
Tooling holes	Added by JLCPCB (the board has none)
Confirm parts placement	**Yes**: check the preview below before production

Upload `jlc_bom.csv` and `jlc_cpl.csv`: 8 part numbers, 263 placements per board (240 LEDs, 19 resistors, 2 capacitors, USB-C, MCU). The test pads and the touch star are bare copper and aren't in the files.

In the placement preview, check:

- **LEDs D1–D240:** each LED's cathode mark must sit on its **lower-left** pad (pad 1, seen from the front). LCSC's footprint for C965810 has the cathode on pad 1, the same as KiCad's, so the 45° rotations in the CPL need no correction. Look at a few LEDs in every row.
- **U1 (PY32F003, TSSOP-20):** the pin-1 dot belongs at the **top-left** corner, next to the ▶ printed on the board. LCSC's footprint for this part is drawn turned 90° compared with KiCad's, which usually means rotation 270 instead of 0. It looked right in the preview on 2026-09-24, so the CPL keeps 0. If it shows turned, fix it in the preview.
- **J1 (USB-C):** the mouth faces the trunk's right edge. Its footprint is LCSC's own, so it lines up.
- Resistors and capacitors aren't polarised.

Stock: one board takes 240 LEDs (XL-1608WWC-06, C965810). Check LCSC has 240 × boards plus JLC's attrition before ordering. LED, MCU and USB-C are *extended* parts (a small fee per part type).

Order 2: the stand fin (PCB only)

Upload `pixelpine_stand_gerbers.zip`. Same settings as the tree, **no assembly**:

Setting	Value
Layers / Dimensions	2 / 36.00 × 10.00 mm (no copper, silkscreen only)
PCB thickness	**1.6 mm**
Surface finish	LeadFree HASL (there's no exposed copper; keeps the RoHS paperwork simple)
Mark on PCB	Remove Mark

Board check (KiCad DRC, 2026-10-02)

- 0 unconnected items, 0 schematic-parity issues, zone fill up to date.
- 2 *starved thermal* errors: J1 pad B12 and R18 pad 2 reach the GND pour through one thermal spoke instead of the required two. Both pads also have their own GND track, so the connection is solid. This is cosmetic, and JLC doesn't check for it. To clear it, give those two pads a solid zone connection in KiCad, or set the zone's minimum spoke count to 1.
- Design rules match JLC's standard 2-layer process (no surcharges): 0.10 mm minimum track and space, 0.3 mm drill / 0.4 mm vias, 0.3 mm copper-to-edge.

Don't run `hardware/scripts/build_all.py` on this board: it regenerates the PCB from the scripts and would discard the hand edits. It now refuses to run unless you pass `--regenerate`.